

Meeting Requirement Specification Report

Generated on: 2026-09-10 15:20:13 | Evaluation Informed by ISO/IEC/IEEE 29148 Principles | Demo Simulation

WITHOUT CLARIFICATION (Baseline)	WITH AI CLARIFICATION (Refined)	MEASURED QUALITY DELTA
8 / 100 Poor (0% Testability)	72 / 100 Excellent (100% Testability)	+64 Points ISO 29148 Score Gain

1. Stakeholder Clarification Q&A; Audit Log

# / Category	Targeted Question & Stakeholder Decision
Q1 [Ranking Algorithm]	Q: How should skills, experience, project complexity, and fresher profiles be weighted? Triggered by: "Mainly skills, experience, and overall profile strength. Things like good companies, solid projects, and impactful work prioritize experience. What? Not strictly, a strong pressure can beat someone with fire average years." Stakeholder Answer: Multi-factor formula: 45% skills match + 35% project complexity + 20% experience with tier normalization
Q2 [Data Ingestion]	Q: What file formats, file size limits, and structured schema validations must be supported? Triggered by: "Do we have historical hiring data to train on?" Stakeholder Answer: Support PDF, DOCX, and scanned PNG/JPG with OCR up to 25MB
Q3 [Explainability]	Q: How should candidate match reasoning and scoring factors be presented to recruiters? Triggered by: "Oh, accurate. Should the system be, and do we need explainability?" Stakeholder Answer: Interactive candidate scorecard showing matched skills %, project impact score, and top 3 justification reasons
Q4 [Accuracy]	Q: What objective accuracy metric defines 'good enough for HR trust' (e.g. Precision@10 >= 85%, NDCG >= 0.82)? Triggered by: "It should be good enough that HR trust fit. And yes, explainability would be useful. For example, why a candidate was ranked higher?" Stakeholder Answer: Ranking Precision@10 >= 85.0% and NDCG@10 >= 0.82 on verified recruiter test set
Q5 [Fairness]	Q: What quantitative fairness metric and audit frequency should be enforced to prevent demographic bias? Triggered by: "Any concerns regarding bias or fairness?" Stakeholder Answer: Disparate Impact Ratio 0.80 - 1.25 across gender and college tiers with automated PII masking
Q6 [Performance]	Q: What specific latency threshold (SLO) defines acceptable performance for resume parsing and ranking? Triggered by: "Real time or batch processing, and what's the timeline?" Stakeholder Answer: Single resume < 1.50s (p95) and batch of 100 resumes < 30.0s

2. Refined Functional Requirements (FR)

ID / Title	Specification & Acceptance Criteria	Status / Source
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FR-01 Resume Ingestion & Parsing Engine	The system shall ingest resumes in PDF and DOCX formats (up to 10MB per file) and parse text into structured JSON schemas as specified by stakeholder: Support PDF, DOCX, and scanned PNG/JPG with OCR up to 25MB. Acceptance Criteria: - Given a valid PDF or DOCX resume under 10MB, When uploaded, Then the system extracts text and generates valid structured JSON. - Given a corrupted or unsupported file type, When uploaded, Then the system returns an informative HTTP 422 error.	RESOLVED STAKEHOLDER_CLARIFICATION
FR-02 Deterministic Multi-Factor Candidate Scoring Engine	The system shall calculate composite candidate relevance scores using stakeholder-defined weighting: Multi-factor formula: 45% skills match + 35% project complexity + 20% experience with tier normalization. Acceptance Criteria: - Given a job description and candidate profile, When scored, Then the final score strictly follows the stakeholder weighting formula. - Given an exceptional fresher with high project complexity, When ranked against generic experience, Then the fresher outranks when composite score is higher.	RESOLVED STAKEHOLDER_CLARIFICATION
FR-03 Structured Explainability & Match Scorecard	The system shall generate an interactive candidate match scorecard satisfying stakeholder preference: Interactive candidate scorecard showing matched skills %, project impact score, and top 3 justification reasons. Acceptance Criteria: - Given a shortlisted candidate, When viewed in the UI, Then the system displays matched skills %, project complexity score, and top 3 justification reasons. - Given recruiter requests match explanation, When requested, Then the explanation payload renders in < 500ms.	RESOLVED STAKEHOLDER_CLARIFICATION
FR-04 Role-Based Recruiter & Hiring Manager Dashboard	The system shall support the recruiter workflow for submitting job descriptions and reviewing candidate shortlists; access-control details require stakeholder clarification. Acceptance Criteria: The recruiter can submit a job description and review a candidate shortlist.	PENDING_CLARIFICATION RAW_DIALOGUE

3. Refined Non-Functional Requirements (NFR)

ID / Category	Requirement Statement & Verifiable Metric	Verification Method
NFR-PERF-01 [Performance]	The system shall meet latency SLO: Single resume < 1.50s (p95) and batch of 100 resumes < 30.0s. Target Metric / SLO: Single resume < 1.50s (p95) and batch of 100 resumes < 30.0s	Automated Load Benchmark (k6/Locust) & Datadog APM Tracing
NFR-FAIR-01 [Fairness]	The ranking algorithm shall enforce demographic fairness standard: Disparate Impact Ratio 0.80 - 1.25 across gender and college tiers with automated PII masking. Target Metric / SLO: Disparate Impact Ratio 0.80 - 1.25 across gender and college tiers with automated PII masking	Automated Fairness Audit Suite (AIF360/Fairlearn) & PII Redaction Test
NFR-ACC-01 [Accuracy]	The candidate shortlisting model shall achieve accuracy benchmark: Ranking Precision@10 >= 85.0% and NDCG@10 >= 0.82 on verified recruiter test set. Target Metric / SLO: Ranking Precision@10 >= 85.0% and NDCG@10 >= 0.82 on verified recruiter test set	Offline Evaluation Benchmark & Human Recruiter Ground-Truth Test Set

NFR-SCOPE-01 [Scope]	The project engineering team shall deliver the validated MVP deliverable compliant with: Single resume < 1.50s (p95) and batch of 100 resumes < 30.0s. Target Metric / SLO: Single resume < 1.50s (p95) and batch of 100 resumes < 30.0s	Sprint Milestone Verification & Release Checklist Sign-off
NFR-SEC-01 [Security]	Candidate personally identifiable information (PII) must be handled securely; the encryption and compliance baseline requires stakeholder clarification. Target Metric / SLO: Unspecified - Awaiting Stakeholder Clarification	Automated Test Suite
NFR-USAB-01 [Usability]	The recruiter dashboard interface shall be usable for daily recruiter tasks; the usability benchmark requires stakeholder clarification. Target Metric / SLO: Unspecified - Awaiting Stakeholder Clarification	Automated Test Suite

4. Quality Comparison Audit Matrix (ISO 29148 Principles)

Quality Dimension	Without Clarification	With Clarification	Improvement (Delta)
Overall Quality Index (OQI)	8/100	72/100	+64 pts
Ambiguity Score	100%	30%	-70% reduction
Testability & Verifiability	0%	70%	+70% gain
Completeness Coverage	0%	69%	+69% gain
Specificity & Measurability	0%	70%	+70% gain
Traceability & Lineage	50%	85%	+35% gain